



DATA ANALYSIS REPORT

Project: Surplus Food Redistribution Network

Team: Catalyst Alliance

UN Sustainable Development Goal: Goal 2 – Zero Hunger

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Status: Final

Last Updated: 11 August 2026

Dataset Overview

Purpose

This document summarizes the exploratory analysis conducted on the operational inventory dataset used to support the Surplus Food Redistribution Network project. The analysis identifies surplus inventory, products at risk of waste, inventory movement patterns, and redistribution opportunities that informed the proposed solution.

Dataset Description

The dataset contains inventory management records, including product categories, suppliers, warehouse locations, inventory quantities, reorder levels, expiration dates, and inventory turnover information.

The analysis was conducted to identify:

- Surplus inventory suitable for redistribution.
- High-risk food categories.
- Products approaching expiry.
- Supplier contribution to overstock.
- Warehouse locations requiring immediate intervention.
- Inventory movement trends.

Methodology Note

Due to the limited availability of publicly accessible retail food waste datasets for African countries, a cleaned, publicly available operational dataset was used to demonstrate the functionality of the proposed solution.

African reports, statistics, and supporting literature were used to establish the problem context and research grounding, while this dataset was used solely to demonstrate how a surplus food redistribution platform could identify, prioritize,

and coordinate food donations.

Executive Summary

Analysis of the inventory dataset indicates that the organization currently holds **27,020 inventory units** with an estimated inventory value of **\$158,777**.

Of this inventory, **2,193 units** were identified as overstocked, representing products that exceed operational demand and may be suitable for redistribution before becoming food waste.

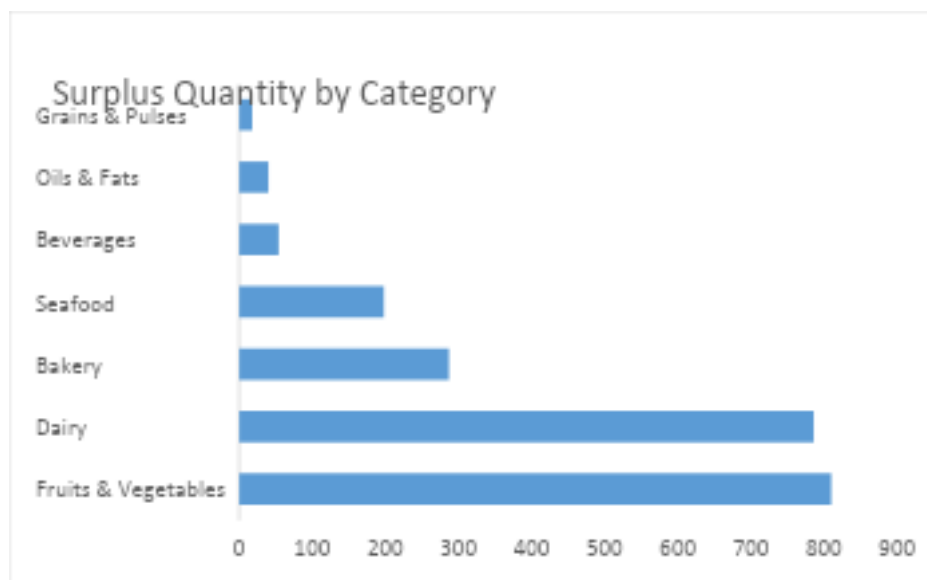
The analysis also identified specific product categories, suppliers, warehouses, and products that contribute most significantly to the risk of food waste.

These findings provide practical evidence supporting the implementation of a data-driven surplus food redistribution platform.

Key Findings

1. Overstock Inventory Presents a Redistribution Opportunity

The analysis identified **2,193 overstocked units**, indicating that a significant portion of inventory exceeds operational requirements.



Rather than allowing these products to remain in storage until demand declines or products expire, they should be prioritized for redistribution to organizations capable of utilizing them.

Project Implication

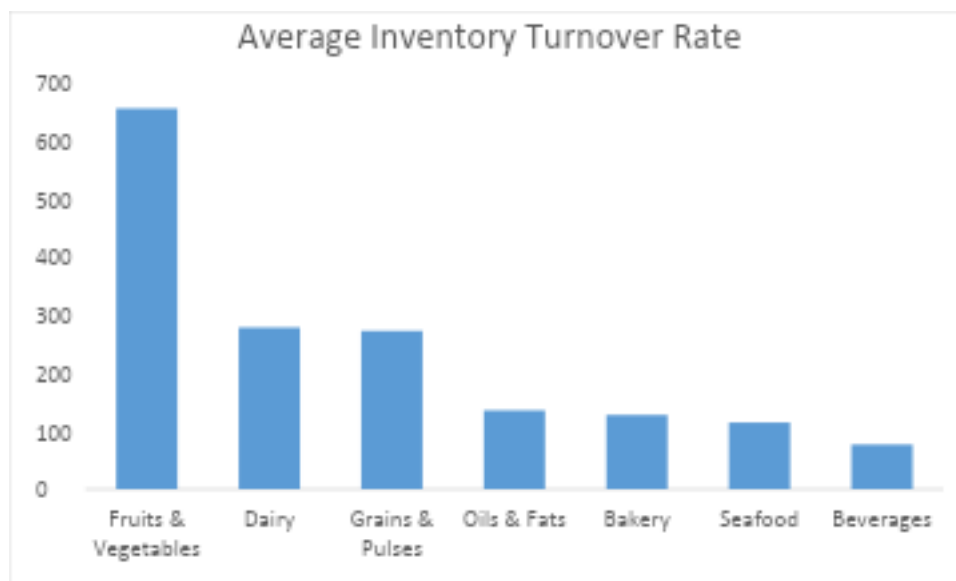
Reducing excess inventory can:

- Minimize food waste.
- Lower storage costs.
- Improve inventory efficiency.
- Increase the quantity of edible food available for redistribution.

2. Fruits and Vegetables Represent the Highest Waste Risk

Among all inventory categories, **Fruits and Vegetables** recorded:

- The highest surplus quantity (**810 units**).
- The lowest average inventory turnover rate.



Because these products are highly perishable, the combination of high surplus inventory and slow turnover significantly increases the likelihood of spoilage.

Project Implication

This category should receive the highest priority within the redistribution platform through:

- Earlier notifications.
- Faster matching with recipient organizations.
- Continuous inventory monitoring.

3. Dairy Products Require Proactive Redistribution

The Dairy category ranked second in both:

- Surplus inventory.

- Slow inventory turnover.

Given the relatively short shelf life of dairy products, prolonged storage increases the risk of spoilage.

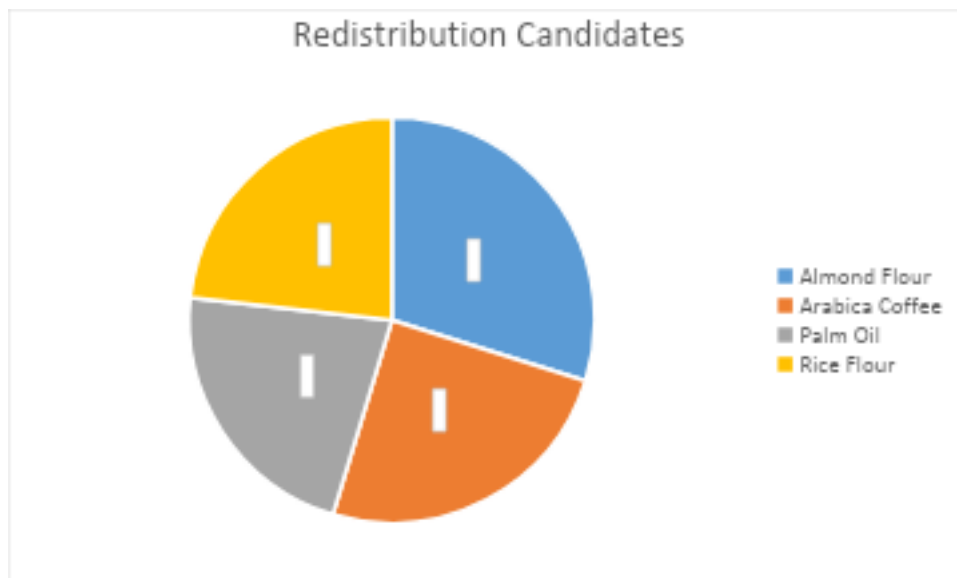
Project Implication

The platform should encourage earlier redistribution of dairy products before they approach expiration.

4. High-Priority Redistribution Products

The analysis identified several products suitable for immediate redistribution:

- Almond Flour (30%)
- Arabica Coffee (25%)
- Rice Flour (23%)
- Palm Oil (22%)



These products satisfy the project's redistribution criteria and should be prioritized before inventory accumulation results in waste.

Project Implication

The platform should highlight high-risk products requiring immediate collection by partner organizations.

5. Warehouse with Highest Waste Risk

The warehouse located at **30880 Barnett Junction** contains the largest concentration of products approaching their expiration dates.



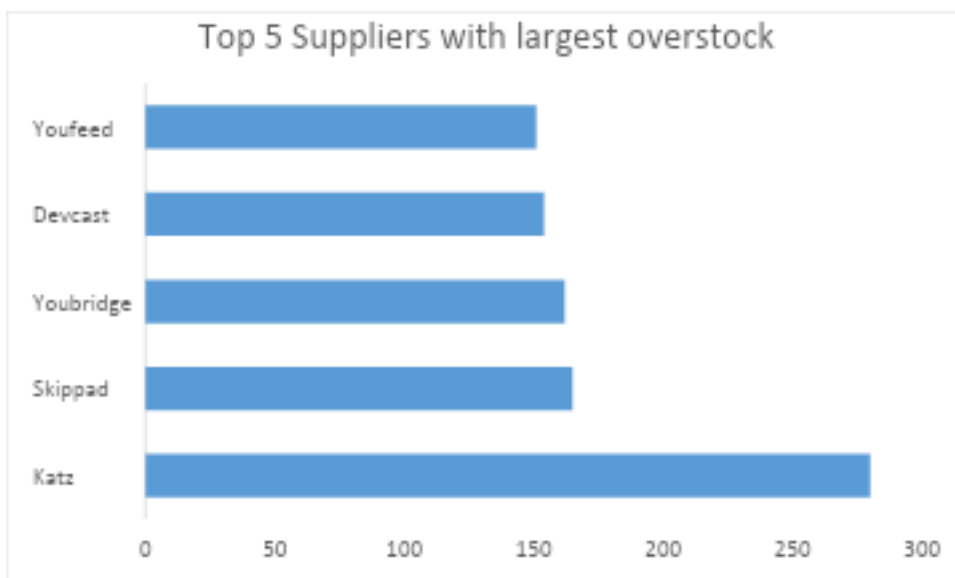
Project Implication

Redistribution activities should prioritize this warehouse to maximize food recovery before products expire.

6. Overstock Concentrated Among Selected Suppliers

Supplier analysis showed that the greatest surplus inventory originated from:

- Katz
- Skippad
- Youbridge



This suggests purchasing quantities from these suppliers may not align with actual demand.

Project Implication

Inventory planning should incorporate supplier performance monitoring to reduce recurring overstock.

7. Slow Inventory Movement Increases Food Waste Risk

Inventory turnover analysis identified the following categories as having the slowest inventory movement:

1. Fruits and Vegetables
2. Dairy
3. Grains and Pulses
4. Oils and Fats
5. Bakery
6. Seafood
7. Beverages

Slow-moving inventory remains in storage longer, increasing the likelihood of expiration.

Project Implication

These categories should receive continuous monitoring and proactive redistribution before products become unsuitable for consumption.

Strategic Recommendations

Based on the analysis, the following actions are recommended:

- Prioritize redistribution of surplus products within the Fruits and Vegetables and Dairy categories.
- Initiate redistribution from the warehouse at **30880 Barnett Junction** to recover products approaching expiry.
- Review procurement practices associated with Katz, Skippad, and Youbridge to reduce recurring overstock.
- Monitor high-priority redistribution products to ensure timely transfer before expiration.
- Implement routine monitoring of inventory turnover and surplus levels to identify redistribution opportunities earlier.

How the Findings Inform the Proposed Solution

The analysis directly informed the design of the proposed Surplus Food Redistribution Network.

The website prototype incorporates features that address the identified challenges, including:

- Vendor surplus food listings.
- Product quantity and expiry tracking.
- Location-based matching between food donors and recipient organizations.
- Prioritization of high-risk food categories.
- Dashboard metrics highlighting surplus inventory and potential food waste.
- Data-driven decision-making to support timely redistribution.

These findings demonstrate how inventory analysis can support earlier intervention, reducing food waste while improving food access.

Conclusion

The analysis demonstrates that inventory data can be used to identify surplus food before it becomes waste and prioritize redistribution based on product category, warehouse location, supplier contribution, and inventory movement.

Although the operational dataset is not Africa-specific, it effectively demonstrates the functionality of the proposed solution.

Combined with African research, statistics, and food security reports, the findings support the implementation of a practical, technology-enabled surplus food redistribution network aligned with **United Nations Sustainable Development Goal 2 – Zero Hunger**.

Limitations

This dataset identifies surplus inventory and products at risk of becoming food waste.

However, it does not contain information relating to:

- Food donations.
- Food pantries.
- NGOs.
- Delivery operations.
- Food recipients.
- Redistribution outcomes.

As a result, the dataset supports inventory identification and prioritization but cannot directly measure the effectiveness of food redistribution activities.

Future implementations would benefit from integrating operational inventory data with real-world donation and recipient datasets to evaluate redistribution outcomes more comprehensively.